

Product, Quotient, and Trigonometric Derivatives

ANSWER KEY



The product rule

- 1 Differentiate $f(x) = (3x^2 - 1)(x^3 + 2x)$ using the product rule.
 $f'(x) = 6x(x^3 + 2x) + (3x^2 - 1)(3x^2 + 2) = 6x^4 + 12x^2 + 9x^4 + 6x^2 - 3x^2 - 2 = 15x^4 + 15x^2 - 2.$
- 2 Differentiate $f(x) = x^2 \sin x$.
 $f'(x) = 2x \sin x + x^2 \cos x.$
- 3 Differentiate $f(x) = e^x \cos x$.
 $f'(x) = e^x \cos x - e^x \sin x = e^x (\cos x - \sin x).$

The quotient rule

- 4 Differentiate $f(x) = \frac{3x-1}{x+4}$.
 $f'(x) = \frac{3(x+4) - (3x-1)(1)}{(x+4)^2} = \frac{3x+12-3x+1}{(x+4)^2} = \frac{13}{(x+4)^2}.$
- 5 Differentiate $f(x) = \frac{x^2}{x^2+1}$.
 $f'(x) = \frac{2x(x^2+1) - x^2(2x)}{(x^2+1)^2} = \frac{2x^3+2x-2x^3}{(x^2+1)^2} = \frac{2x}{(x^2+1)^2}.$
- 6 Differentiate $f(x) = \frac{\sin x}{x}$.
 $f'(x) = \frac{x \cos x - \sin x}{x^2}.$

Derivatives of the six trig functions

- 7 State the derivative of each trig function:

a $\frac{d}{dx} \sin x = \cos x$	b $\frac{d}{dx} \cos x = -\sin x$	c $\frac{d}{dx} \tan x = \sec^2 x$
d $\frac{d}{dx} \cot x = -\csc^2 x$	e $\frac{d}{dx} \sec x = \sec x \tan x$	f $\frac{d}{dx} \csc x = -\csc x \cot x$
- 8 Differentiate $f(x) = x \tan x$.
 $f'(x) = \tan x + x \sec^2 x.$
- 9 Differentiate $f(x) = \sec x \cdot \csc x$.
 $f'(x) = \sec x \tan x \csc x + \sec x (-\csc x \cot x) = \sec x \csc x (\tan x - \cot x).$

Combining product, quotient, and trig rules

10 Differentiate $f(x) = \frac{e^x}{\sin x}$.

$$f'(x) = \frac{e^x \sin x - e^x \cos x}{\sin^2 x} = \frac{e^x(\sin x - \cos x)}{\sin^2 x}.$$

- 11 Find the equation of the tangent line to $f(x) = x \cos x$ at $x = 0$.
 $f(0) = 0$. $f'(x) = \cos x - x \sin x$; $f'(0) = 1$. Tangent: $y - 0 = 1(x - 0)$, i.e. $y = x$.

Derivatives using a table of values

- 12 Given $f(3) = 2$, $f'(3) = -1$, $g(3) = 4$, $g'(3) = 5$, find $h'(3)$ if $h(x) = f(x)g(x)$.
 $h'(x) = f'(x)g(x) + f(x)g'(x)$; $h'(3) = (-1)(4) + (2)(5) = -4 + 10 = 6$.

- 13 Using the same table, find $k'(3)$ if $k(x) = \frac{f(x)}{g(x)}$.
 $k'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$; $k'(3) = \frac{(-1)(4) - (2)(5)}{4^2} = \frac{-4 - 10}{16} = -\frac{14}{16} = -\frac{7}{8}$.

Second derivatives with product and quotient rules

- 14 Find $f''(x)$ for $f(x) = x \sin x$.
 $f'(x) = \sin x + x \cos x$. $f''(x) = \cos x + \cos x - x \sin x = 2 \cos x - x \sin x$.

More product and quotient rule practice

- 15 Differentiate $f(x) = (2x + 1)^2(x - 3)$, treating $(2x + 1)^2$ as one factor.
Let $u = (2x + 1)^2$, $u' = 2(2x + 1)(2) = 4(2x + 1)$:
 $f'(x) = 4(2x + 1)(x - 3) + (2x + 1)^2(1) = (2x + 1)[4(x - 3) + (2x + 1)] = (2x + 1)(6x - 11)$.

- 16 Differentiate $f(x) = \frac{x^2 + 1}{x - 1}$.

$$f'(x) = \frac{2x(x - 1) - (x^2 + 1)(1)}{(x - 1)^2} = \frac{2x^2 - 2x - x^2 - 1}{(x - 1)^2} = \frac{x^2 - 2x - 1}{(x - 1)^2}.$$

An applied product-rule problem

- 17 A company's revenue is $R(t) = p(t) \cdot q(t)$, where $p(t)$ is price and $q(t)$ is quantity sold, both functions of time t (months). At $t = 6$, $p(6) = 20$, $p'(6) = 0.5$, $q(6) = 100$, $q'(6) = -2$. Find $R'(6)$ and interpret its sign.
 $R'(t) = p'(t)q(t) + p(t)q'(t)$: $R'(6) = (0.5)(100) + (20)(-2) = 50 - 40 = 10$. Since $R'(6) > 0$, revenue is increasing at $t = 6$ despite falling quantity, because the price increase outweighs the quantity decrease.

Higher-order trigonometric derivatives

- 18** Find $f''(x)$ for $f(x) = \sin x$, and describe the repeating pattern of derivatives f, f', f'', f''', f'''' .
 $f'(x) = \cos x$, $f''(x) = -\sin x$. The pattern $\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x$ repeats every 4 derivatives.
- 19** Using the repeating pattern above, find $f^{(50)}(x)$ for $f(x) = \sin x$ (the 50th derivative).
 Since the pattern has period 4, and $50 = 4(12) + 2$, the 50th derivative matches the 2nd derivative in the cycle: $f^{(50)}(x) = -\sin x$.

A capstone combined-rule problem

- 20** Differentiate $f(x) = \frac{x^2 \sin x}{x+1}$, applying the quotient rule with the product rule for the numerator's derivative.
 Numerator $u = x^2 \sin x$: $u' = 2x \sin x + x^2 \cos x$. Denominator $v = x + 1$: $v' = 1$.

$$f'(x) = \frac{(2x \sin x + x^2 \cos x)(x+1) - x^2 \sin x(1)}{(x+1)^2}.$$
- 21** An object's position is $s(t) = t^2 \cos t$ (meters, seconds). Find the object's velocity function $s'(t)$ using the product rule, then find its velocity at $t = 0$.
 $s'(t) = 2t \cos t - t^2 \sin t$. At $t = 0$: $s'(0) = 2(0)(1) - 0 = 0$ m/s (momentarily at rest).
- 22** Differentiate $f(x) = \frac{\cos x}{1 + \sin x}$, and simplify your answer using the Pythagorean identity $\sin^2 x + \cos^2 x = 1$.

$$f'(x) = \frac{-\sin x(1 + \sin x) - \cos x(\cos x)}{(1 + \sin x)^2} = \frac{-\sin x - \sin^2 x - \cos^2 x}{(1 + \sin x)^2} = \frac{-\sin x - 1}{(1 + \sin x)^2} = \frac{-1}{1 + \sin x}.$$
- 23** Given $f(2) = 5$, $f'(2) = 3$, $g(2) = -1$, $g'(2) = 4$, find each of the following at $x = 2$:
- a** $\frac{d}{dx}[f(x)g(x)]$ $f'(2)g(2) + f(2)g'(2) = (3)(-1) + (5)(4) = -3 + 20 = 17$.
- b** $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right]$ $\frac{f'(2)g(2) - f(2)g'(2)}{[g(2)]^2} = \frac{(3)(-1) - (5)(4)}{1} = \frac{-3 - 20}{1} = -23$.
- c** $\frac{d}{dx}[3f(x) - 2g(x)]$ $3f'(2) - 2g'(2) = 3(3) - 2(4) = 9 - 8 = 1$.